

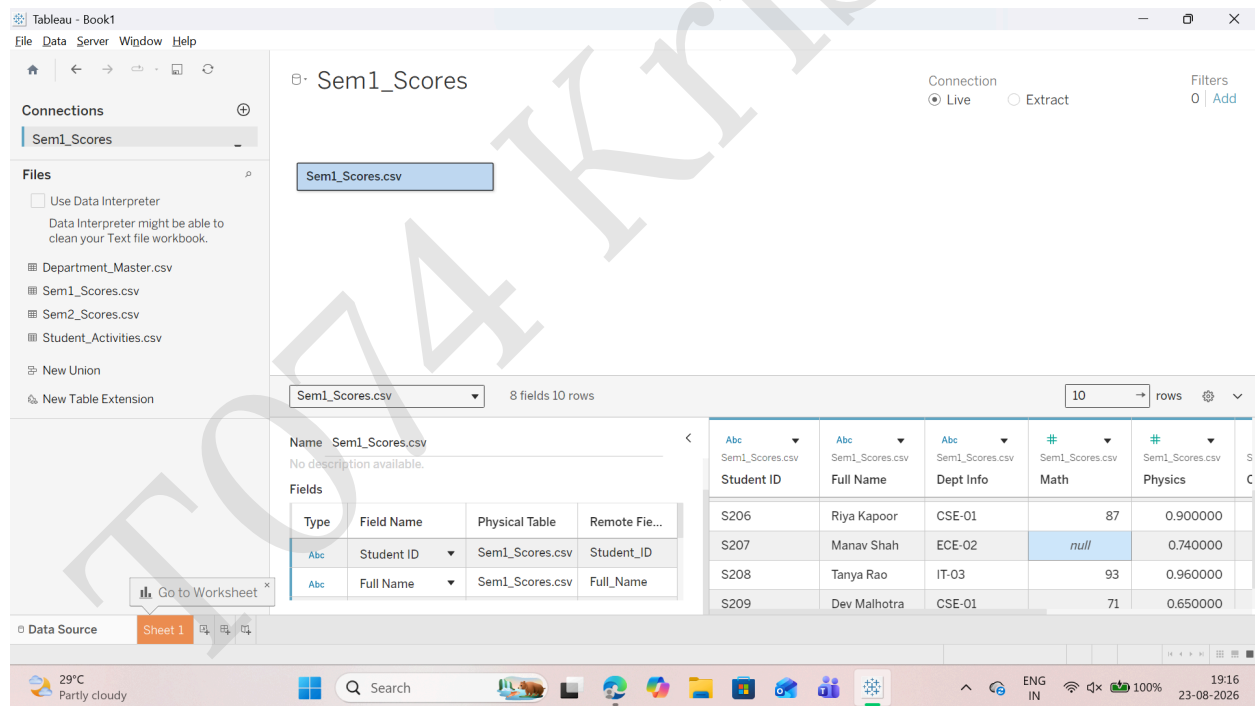
SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU
PRACTICAL - 9

Aim: Cleaning and Structuring Data for Analysis

- a) Identify and handle missing or inconsistent data
- b) Convert wide data into tall format
- c) Perform union of multiple files
- d) Execute cross-database joins
- e) Restructure data for better visualization
- f) Handle different levels of detail in datasets
- g) Apply advanced data cleaning techniques

A: Identify and Handle Missing or Inconsistent Data

Open **Sem1_Scores.csv** in Tableau using **Text File**, then check the Data Source preview to identify **Null values in Math, Physics, and Chemistry**.



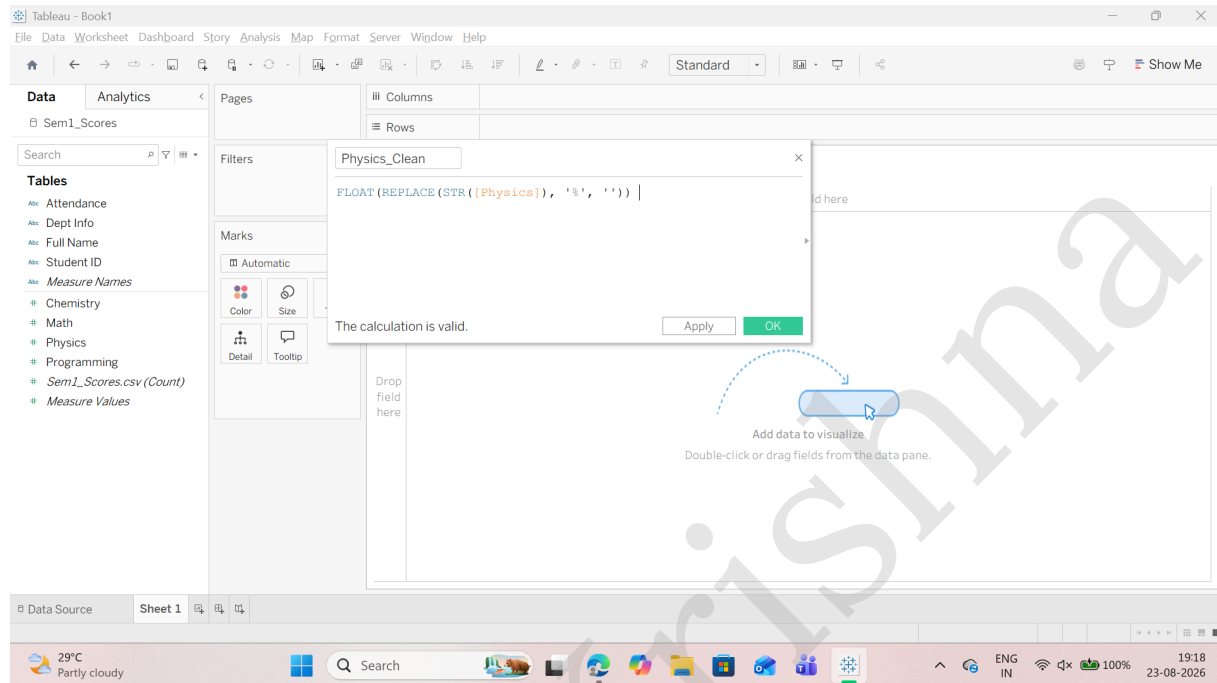
The screenshot shows the Tableau interface with the 'Sem1_Scores.csv' data source loaded. The 'Fields' pane on the left lists the available fields: Student ID, Full Name, Dept Info, Math, and Physics. The 'Data Source' preview table is displayed below the fields pane, showing 8 fields and 10 rows. The table contains the following data:

Student ID	Full Name	Dept Info	Math	Physics
S206	Riya Kapoor	CSE-01	87	0.900000
S207	Manav Shah	ECE-02	null	0.740000
S208	Tanya Rao	IT-03	93	0.960000
S209	Dev Malhotra	CSE-01	71	0.650000

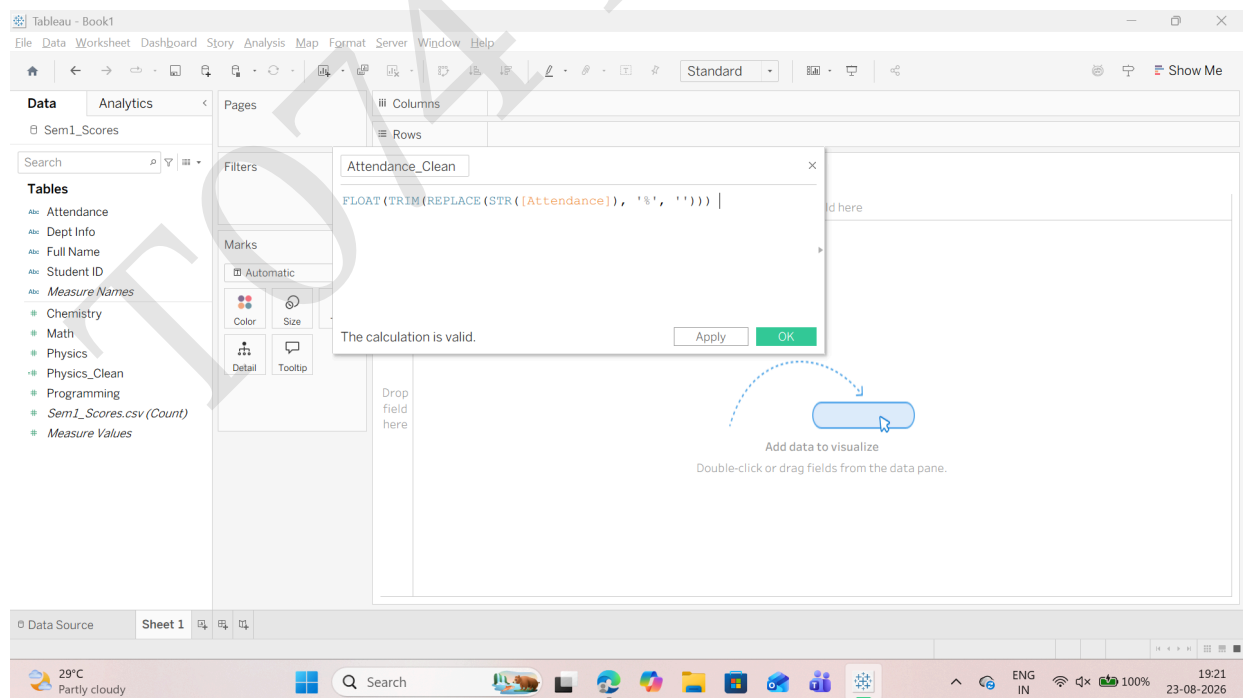
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In **Sheet 1**, create a calculated field named **Physics_Clean** using **FLOAT(REPLACE(STR([Physics]), '%', ''))** to convert Physics text percentages into numeric values.



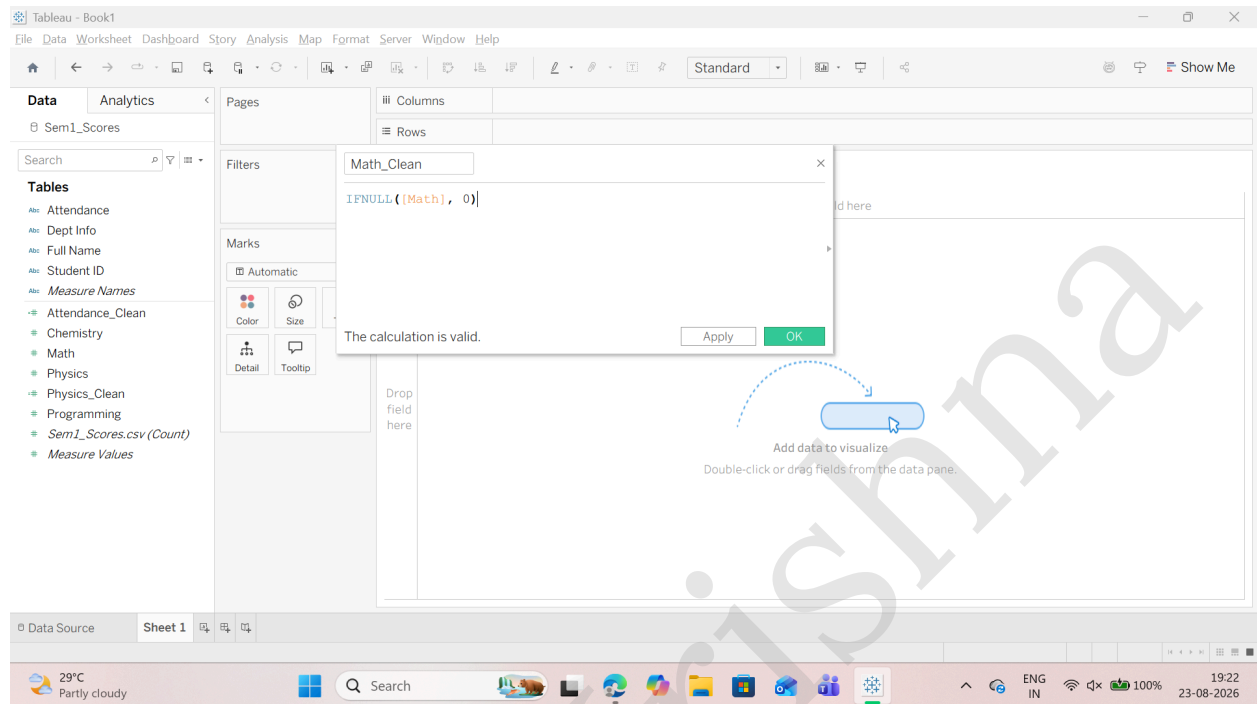
Create **Attendance_Clean** with **FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))** to convert Attendance percentages into numeric values, then click **OK**.



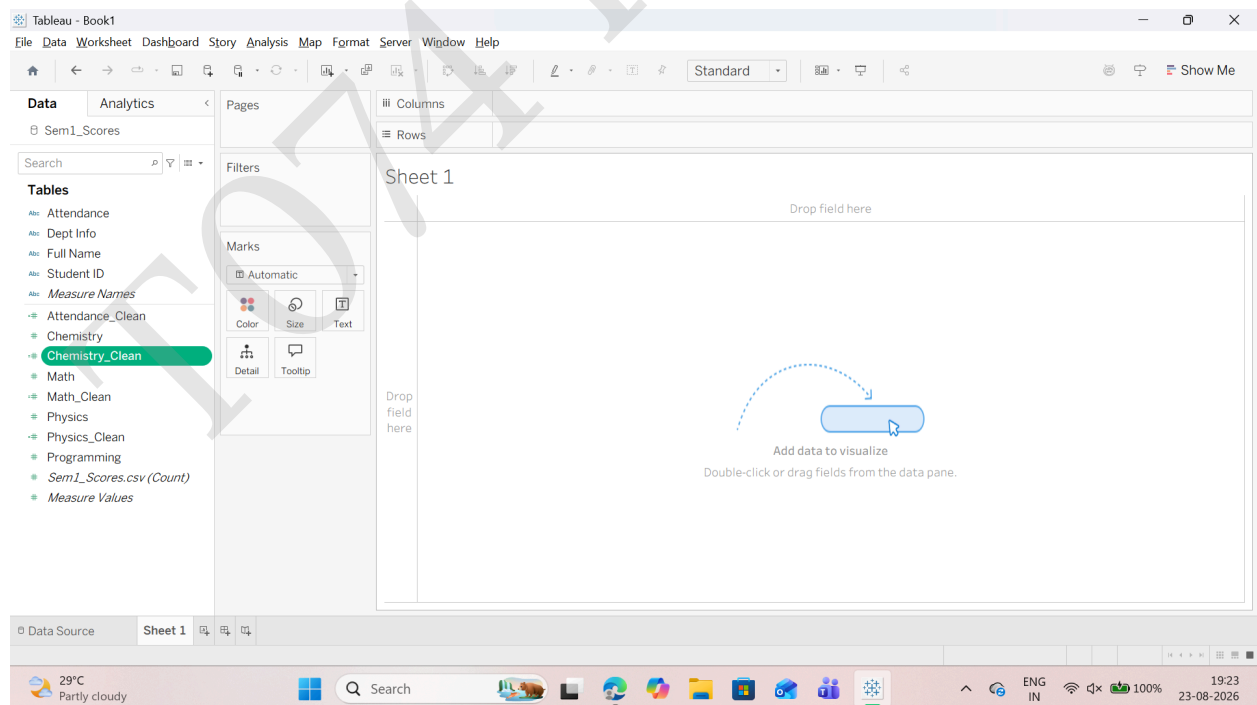
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Create a calculated field named **Math_Clean** using **IFNULL([Math], 0)** to replace missing Math values with **0**, then click **OK**.



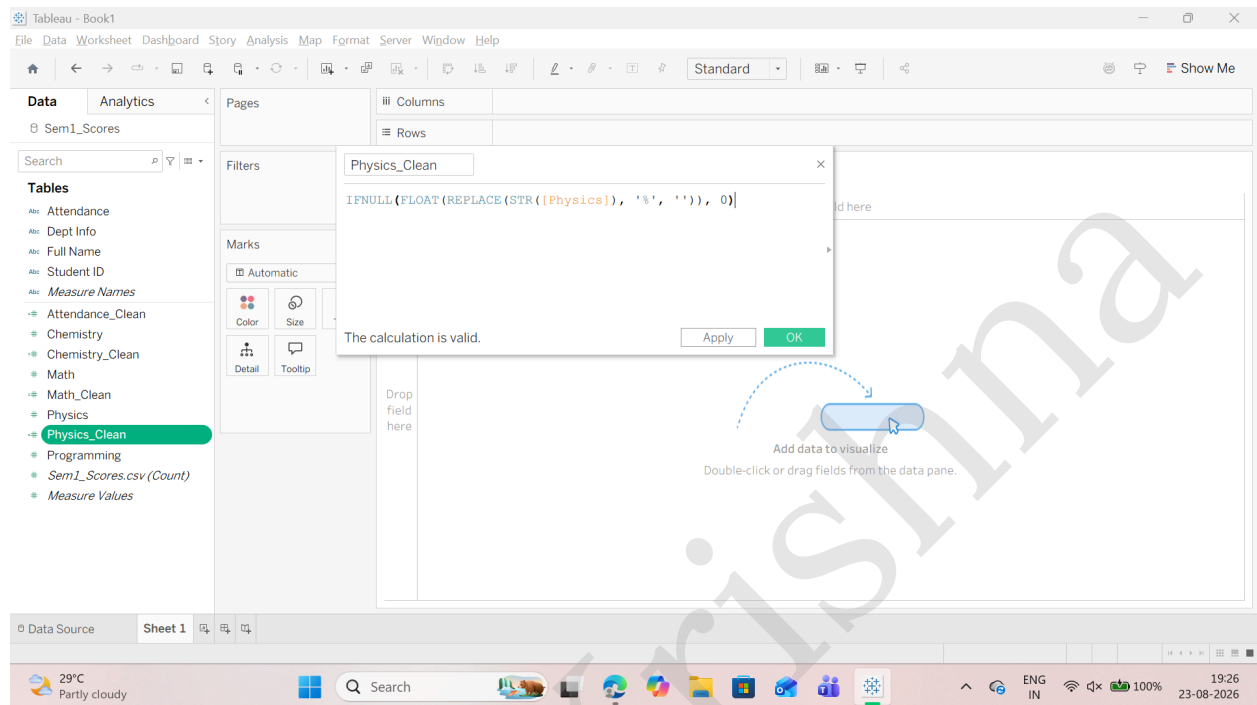
Right-click Data Pane → Create Calculated Field → Name: Chemistry_Clean → Formula: `IFNULL([Chemistry], 0)` → Click OK.



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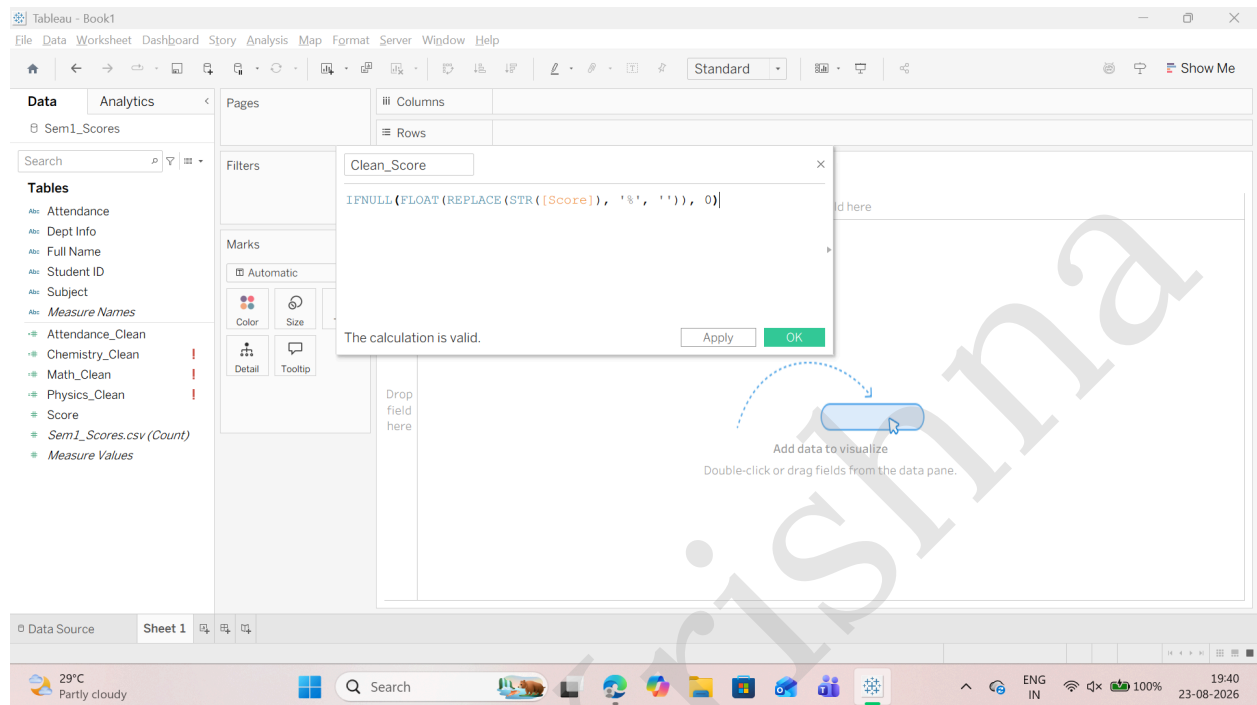
Edit **Physics_Clean** to `IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)` so the Null Physics value for **S105** is replaced with **0**.



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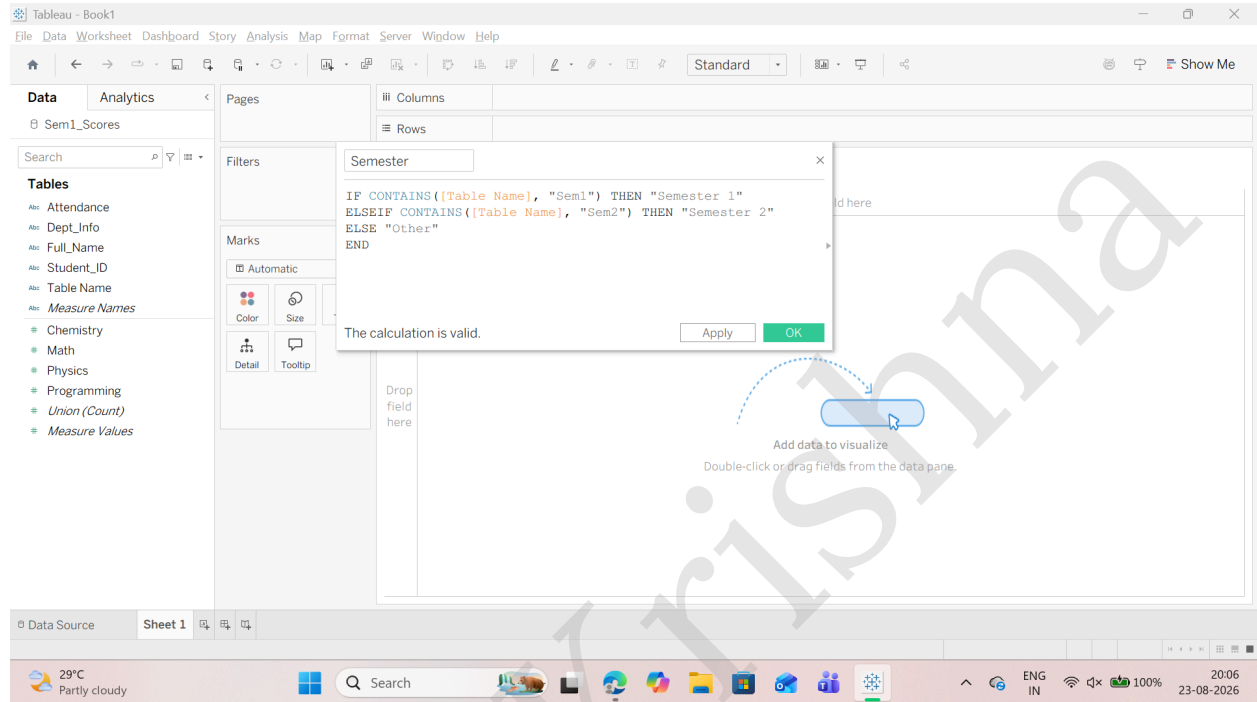
Convert **Score** from text to **Number (decimal)** using the **Abc** dropdown, or create **Clean_Score** with `IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)`.



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Create a calculated field named **Semester** using `IF CONTAINS([Table Name], "Sem1") THEN "Semester 1" ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2" ELSE "Other" END` to identify each semester.



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D: Execute Cross-Database Joins

Stay on the **Top Canvas**, drag **Department_Master.csv** to the right of the Union box, and set the relationship **Dept_Info = Dept_Code**.

Tableau - Book1

File Data Server Window Help

Connections: Sem1_Scores

Files: Department_Master.csv, Sem1_Scores.csv, Sem2_Scores.csv, Student_Activities.csv

Sem1_Scores

Connection: Live

Filters: 0 | Add

Union

Department_Master.csv

Union — Department...

Name	Department_Master.csv
Dept Code	Department Name
CSE-01	Computer Science
ECE-02	Electronics & Communication
IT-03	Information Technology
MECH-04	Mechanical Engineering

Performance Options

Remove Relationship

Drag **Student_Activities.csv** to the right of the Union box and set the relationship **Student_ID = Student_ID** between the Union and Student_Activities tables.

Tableau - Book1

File Data Server Window Help

Connections: Sem1_Scores

Files: Department_Master.csv, Sem1_Scores.csv, Sem2_Scores.csv, Student_Activities.csv

Sem1_Scores

Connection: Live

Filters: 0 | Add

Union

Student_Activities.csv

Union — Student_A...

Name	Student_Activities.csv
Student ID (Student Activ	Club Membership
S201	AI Club
S202	Robotics Club
S203	Coding Club
S204	Literary Society
S205	Sports Club
S206	AI Club
S207	Robotics Club
S208	Literary Society
S209	Sports Club

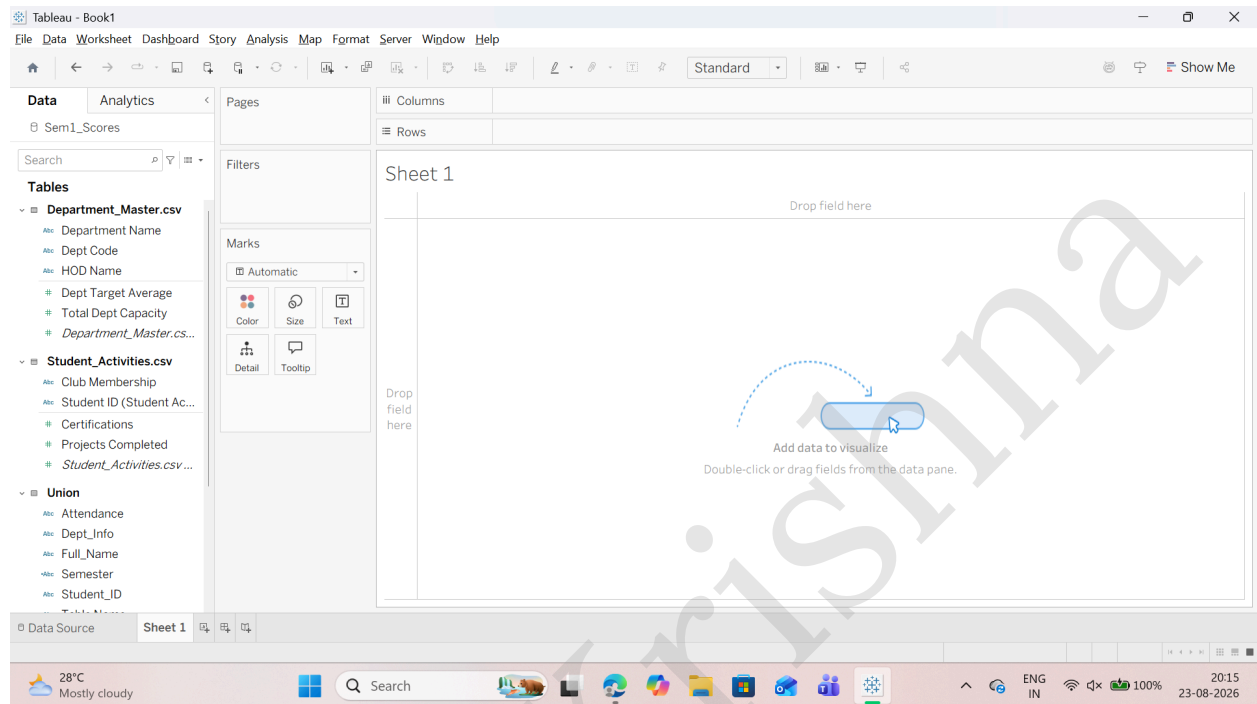
Performance Options

Remove Relationship

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Open **Sheet 1** and verify that the Data Pane contains the fields from **Union**, **Department_Master**, and **Student_Activities**, confirming that all three tables are correctly linked.

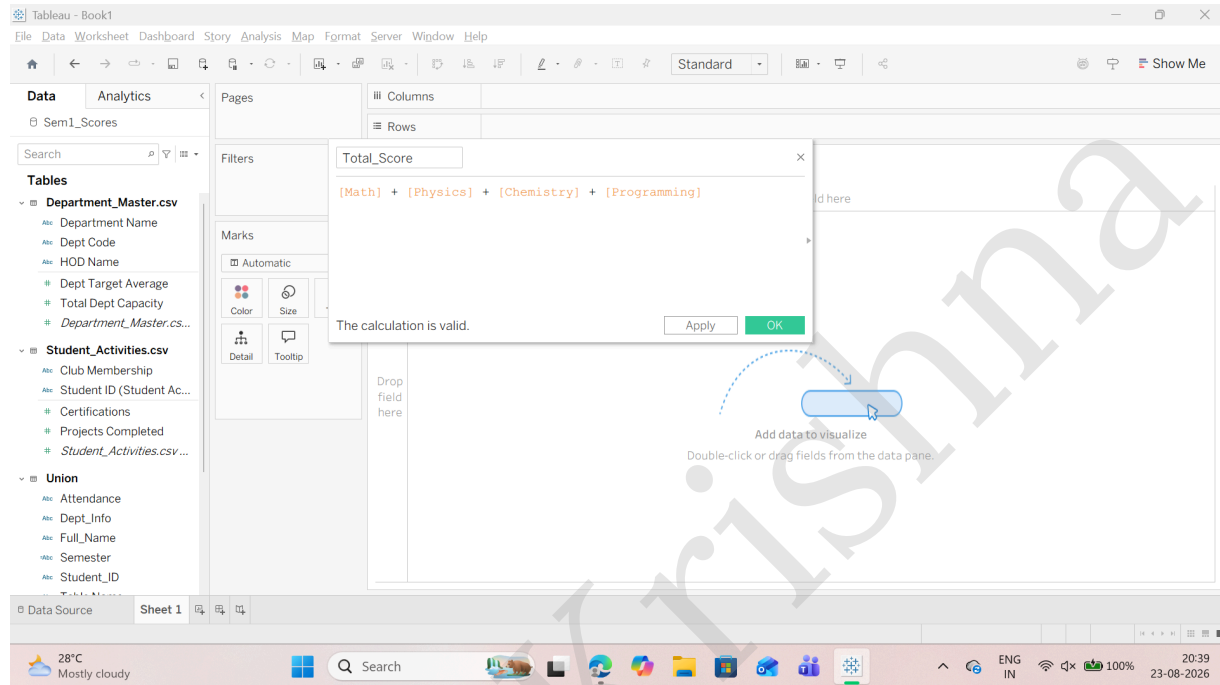


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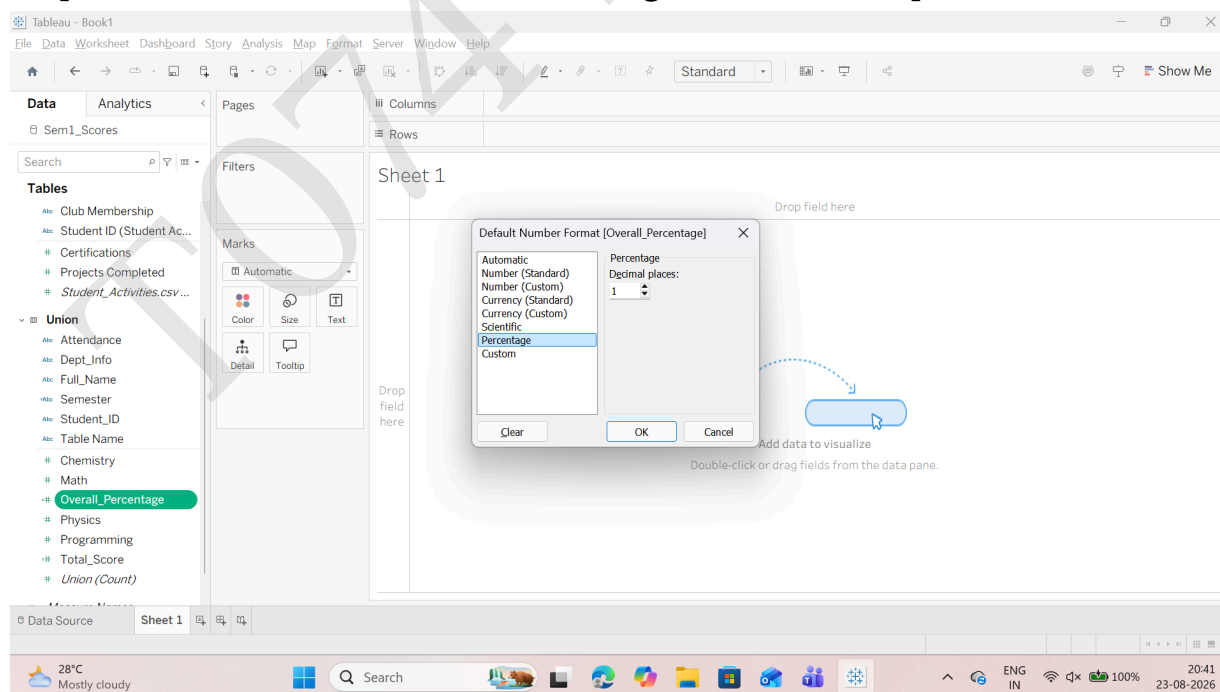
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E: Restructure Data for Better Visualization

Create a calculated field named **Total_Score** using **[Math_Clean]** + **[Physics_Clean]** + **[Chemistry_Clean]** + **[Programming]** to calculate each student's total marks.



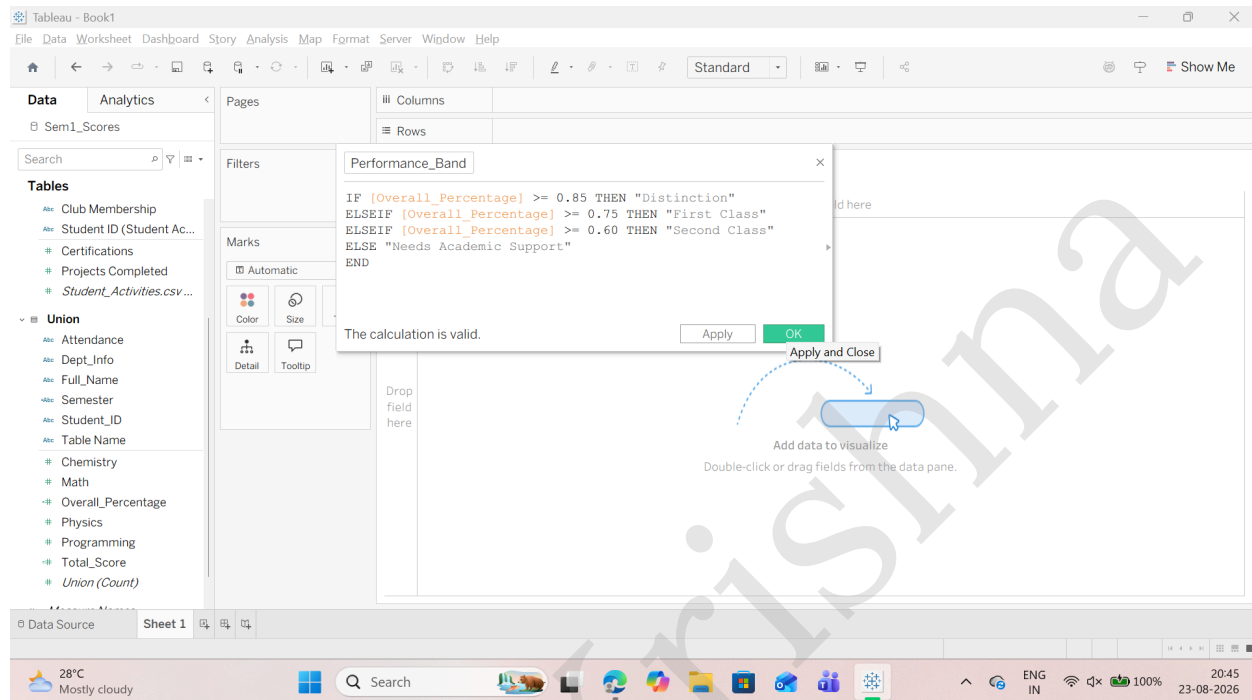
Create **Overall_Percentage** using $[Total_Score] / 400.0$, then set its **Default Properties** → **Number Format** to **Percentage** with **1 decimal place**



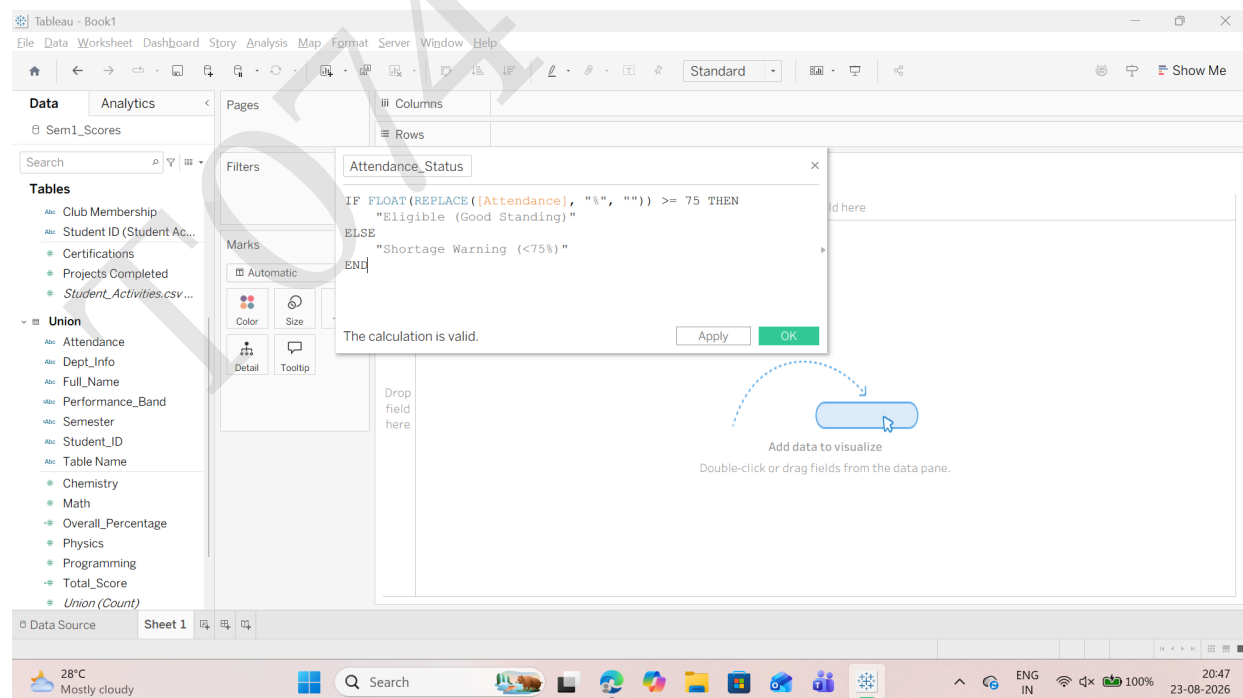
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Create a calculated field named **Performance_Band** using the given percentage thresholds to classify students as **Distinction**, **First Class**, **Second Class**, or **Needs Academic Support**.



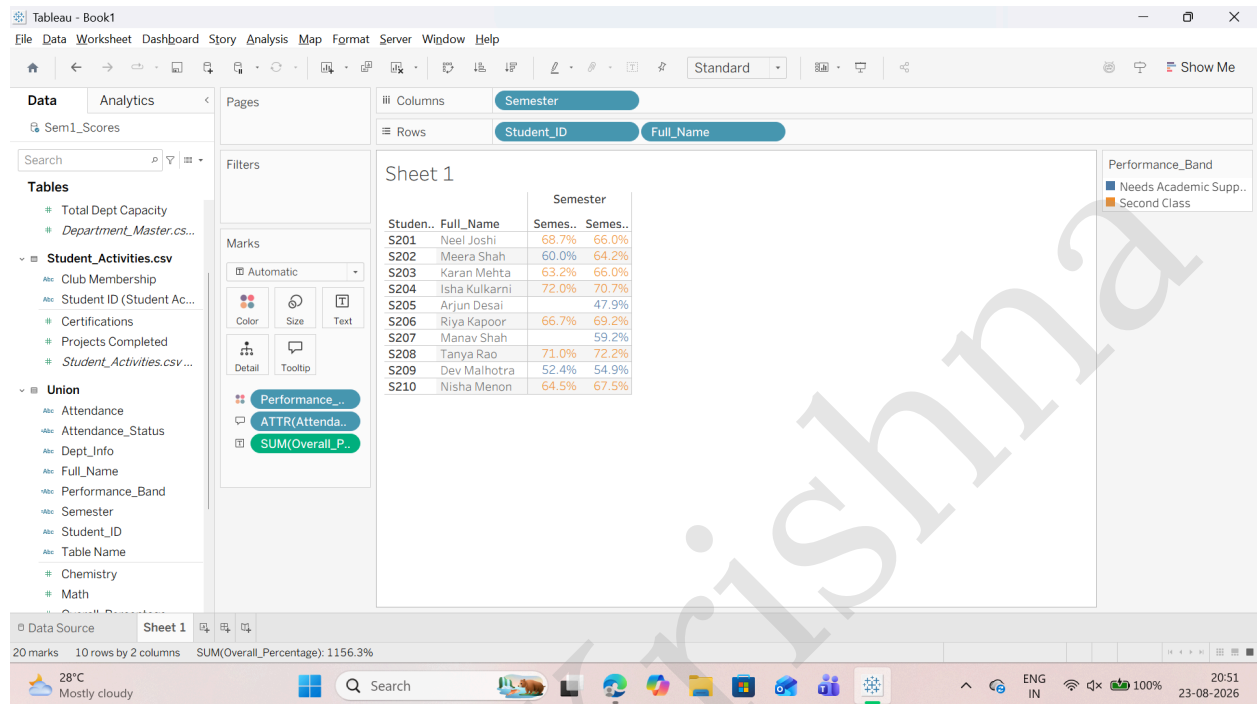
Create a calculated field named **Attendance_Status** using IF
[Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE
"Shortage Warning (<75%)" END.



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Drag **Student_ID** and **Full_Name** to Rows, **Semester** to Columns, **Overall_Percentage** to Text, **Performance_Band** to Color, and **Attendance_Status** to Tooltip or Detail on the Marks card.

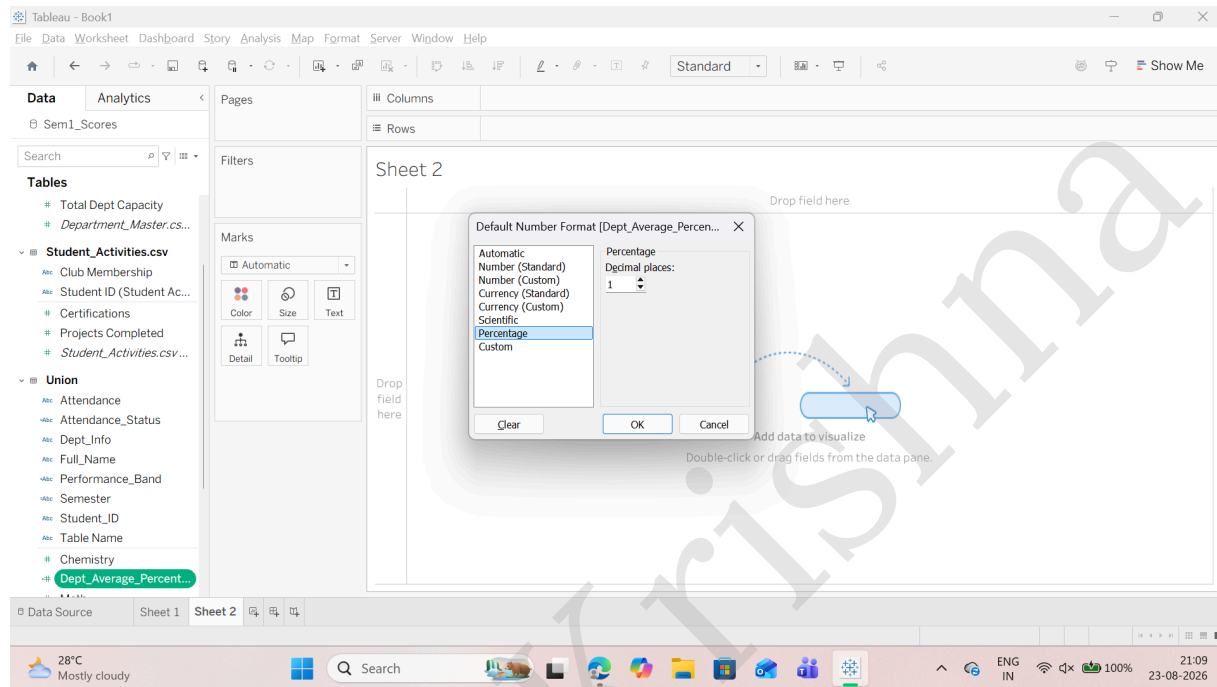


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F: Handle Different Levels of Detail in Datasets

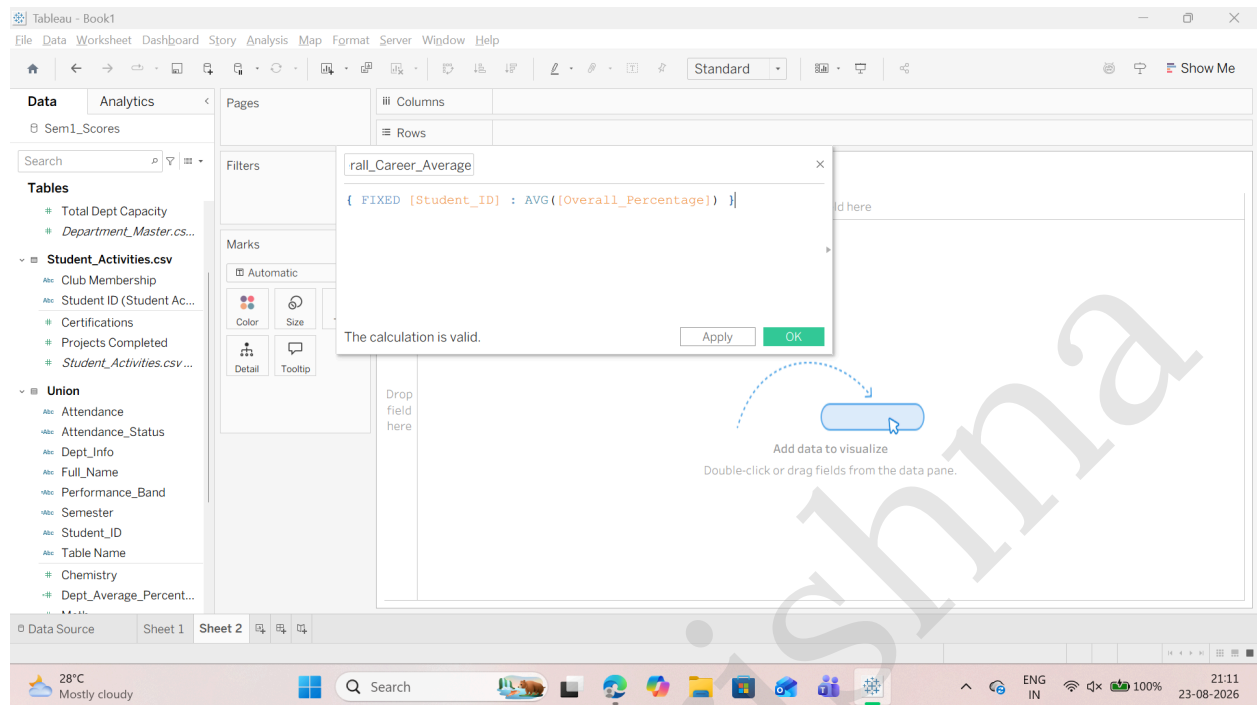
Create **Dept_Average_Percentage_LOD** using { **FIXED [Dept_Info] : AVG([Overall_Percentage])** }, then format it as **Percentage with 1 decimal place**.



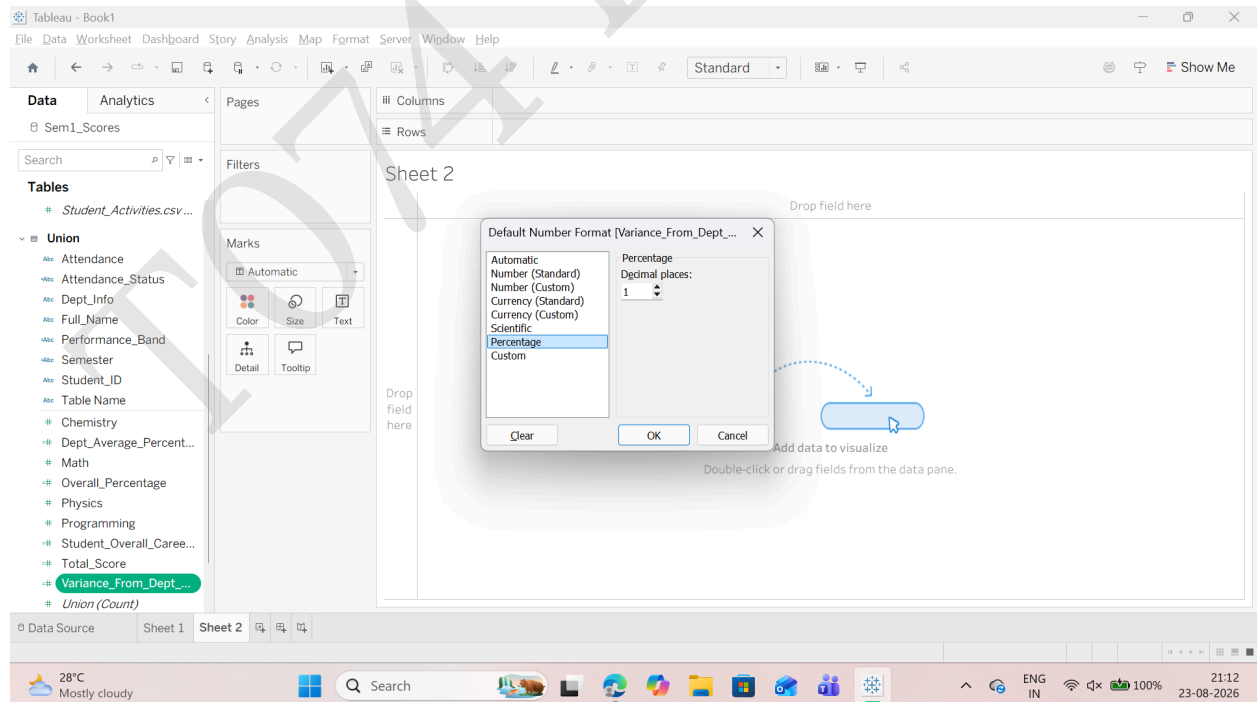
Create **Student_Overall_Career_Average** using { **FIXED [Student_ID] : AVG([Overall_Percentage])** }, then format it as **Percentage with 1 decimal place**.

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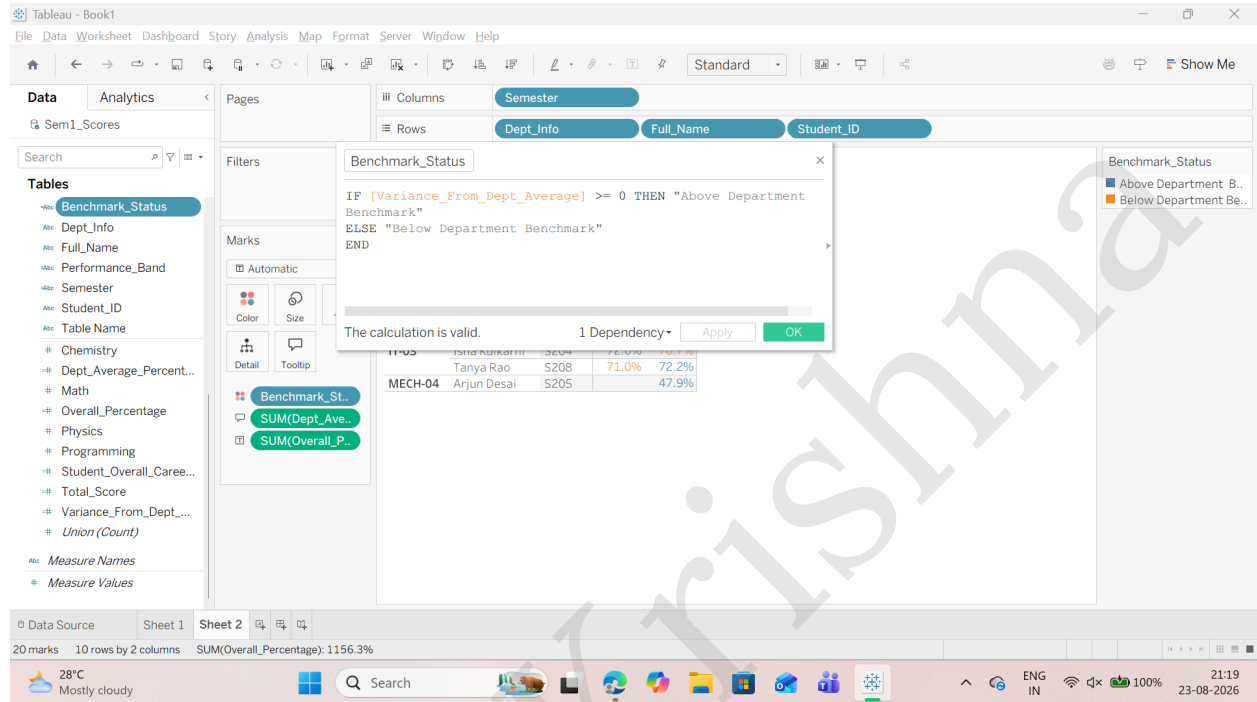
Create **Variance_From_Dept_Average** using `[Overall_Percentage]` - `[Dept_Average_Percentage_LOD]`, then format it as **Percentage with 1 decimal place**.



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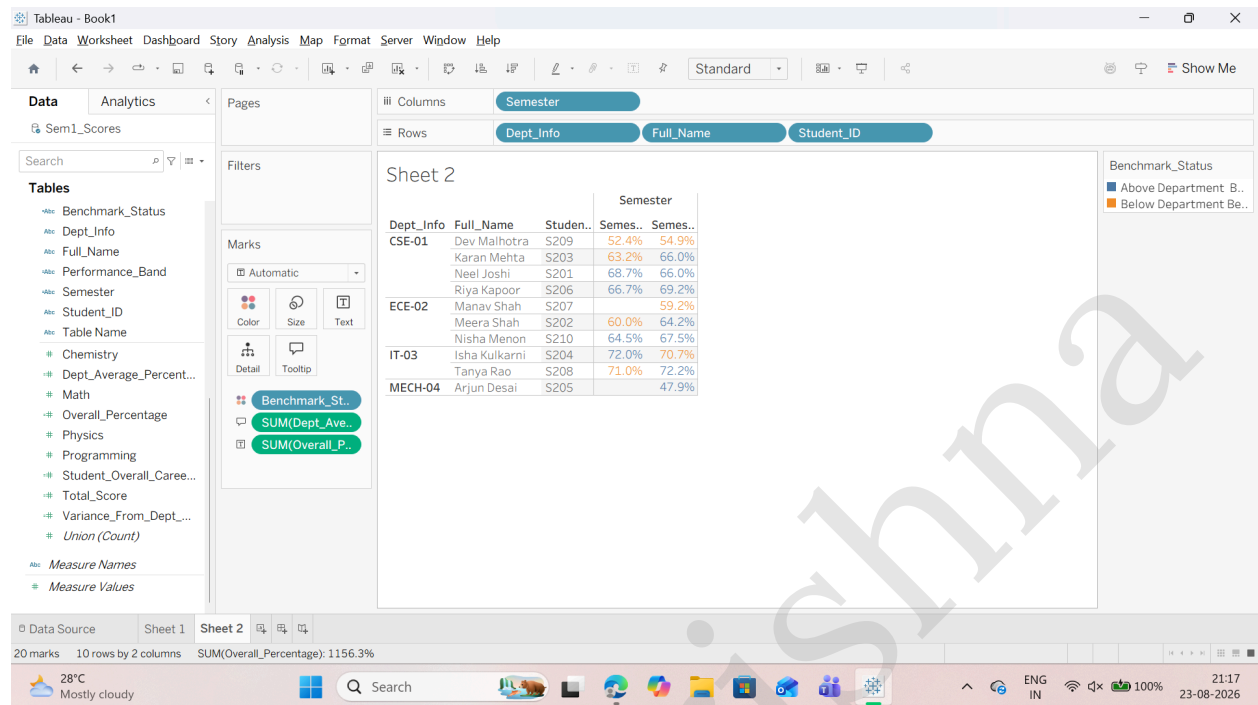
Create **Benchmark_Status** using **IF [Variance_From_Dept_Average] >= 0**
THEN "Above Department Benchmark" ELSE "Below Department
Benchmark" END to classify students against the department benchmark.



Place **Dept_Info**, **Student_ID**, and **Full_Name** on Rows, **Semester** on Columns, **Overall_Percentage** on Text, **Dept_Average_Percentage_LOD** on Tooltip, and **Benchmark_Status** on Color.

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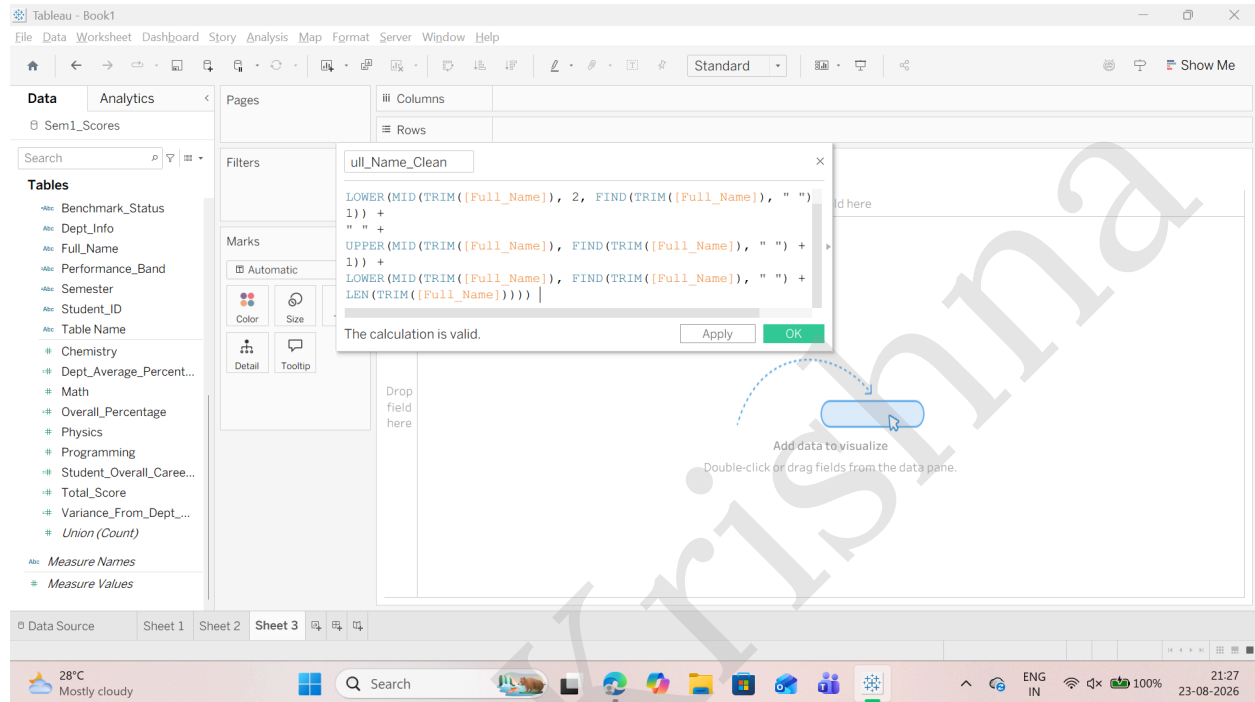


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G: Apply Advanced Data Cleaning Techniques

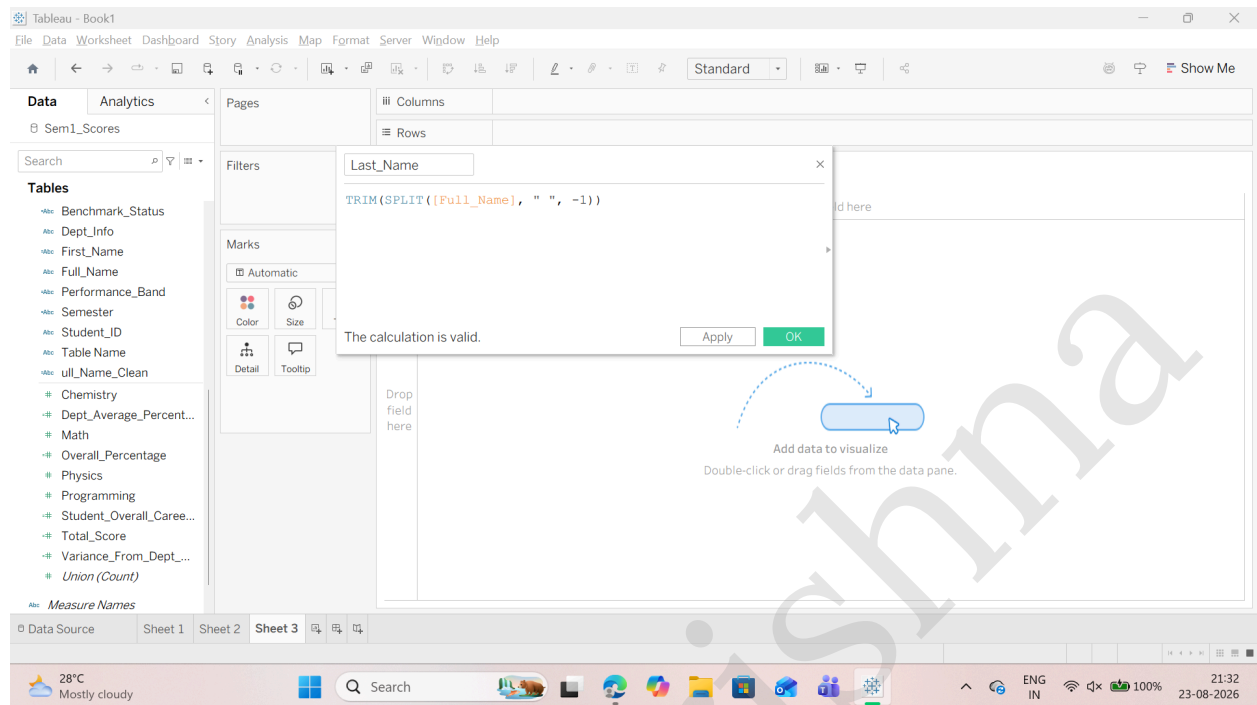
Create **Full_Name_Clean** using `TRIM([Full_Name])` for simple whitespace cleaning, or use the provided formula to standardize the name's capitalization, then click **OK**.



Create **First_Name** using `TRIM(SPLIT([Full_Name], " ", 1))` and **Last_Name** using `TRIM(SPLIT([Full_Name], " ", -1))`.

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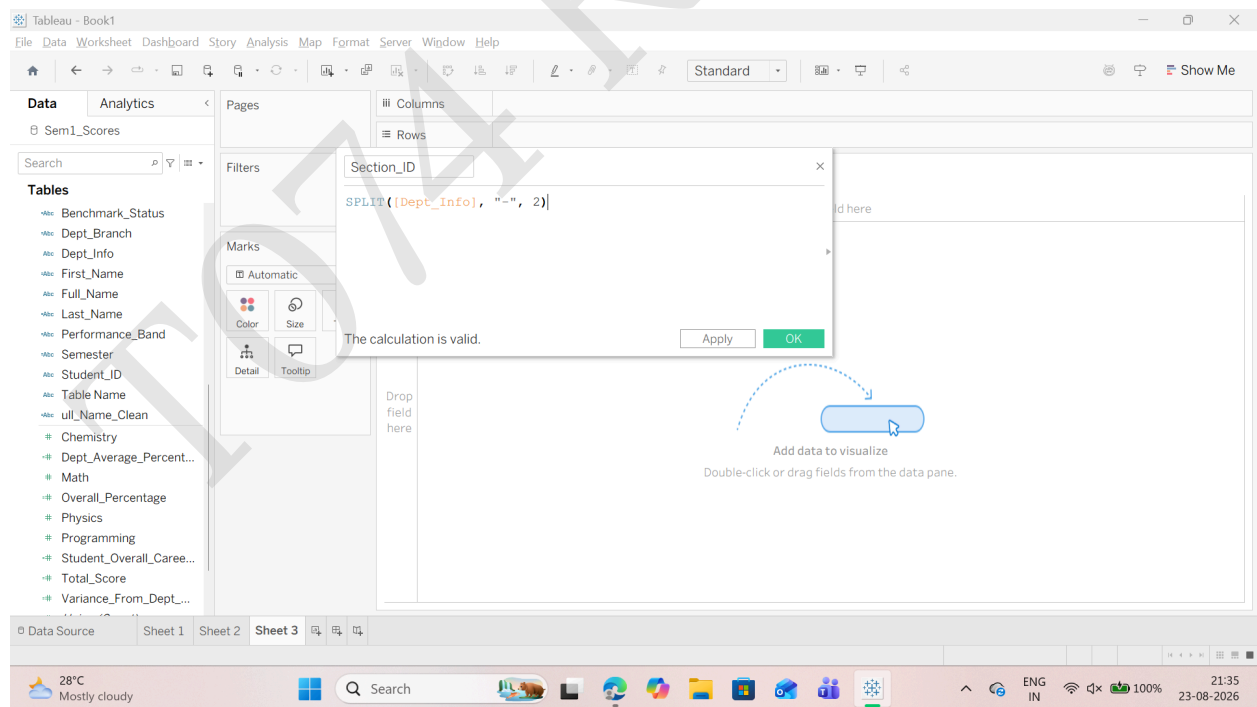
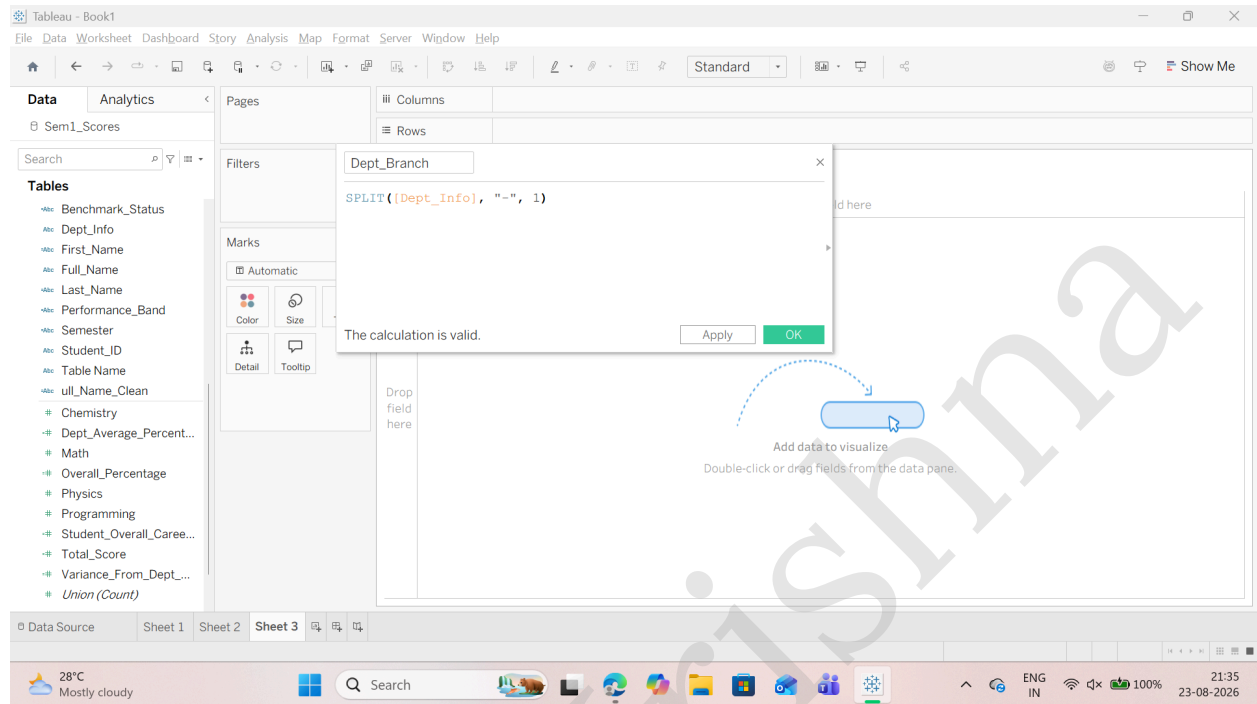
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Create **Dept_Branch** using `SPLIT([Dept_Info], "-", 1)` and **Section_ID** using `SPLIT([Dept_Info], "-", 2)` to separate the department acronym and batch/section ID.

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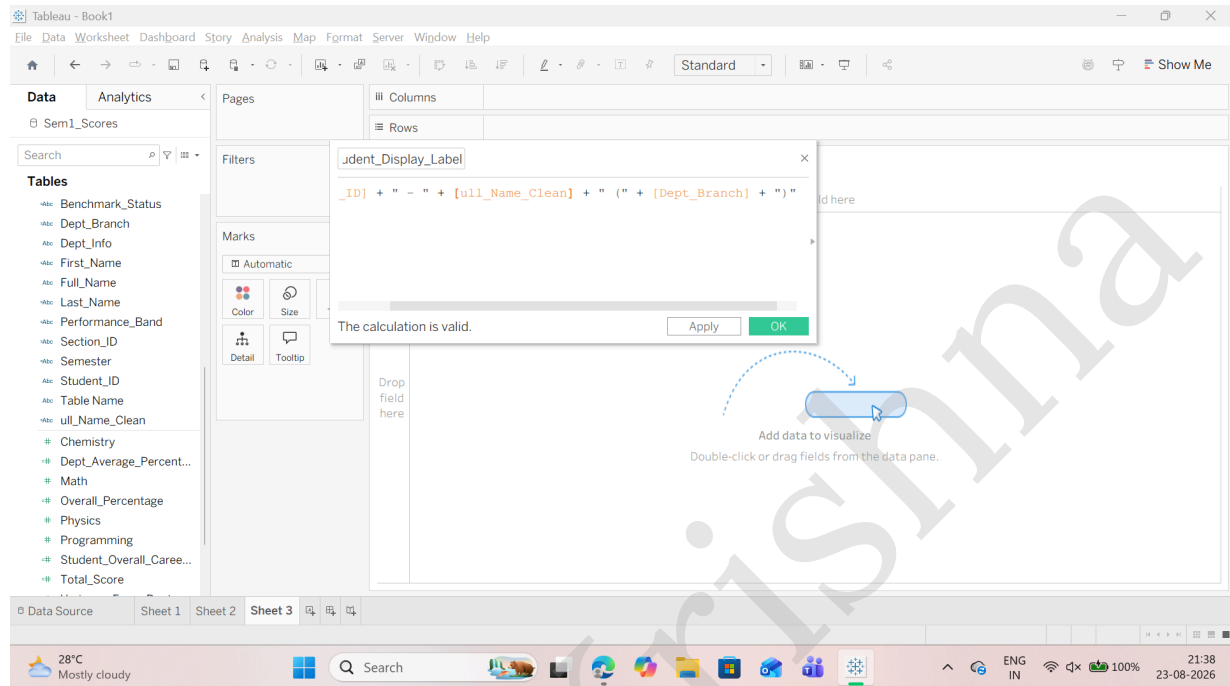
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Create **Student_Display_Label** using `[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"` to combine the student ID, cleaned name, and department code into one label.



Place **Student_Display_Label** on Rows, **Semester** on Columns, **Overall_Percentage** on Text, **Performance_Band** on Color, and **Dept_Branch** and **Section_ID** on Filters.

